

Date:

EXPERIMENT NO. 2**AIM: Understand and identify header fields of layers of TCP/IP protocol stack.****Exercise-1: Input Sequence TCP**

```

111010001101100011010001010001101111001111110001000000000001101010001100011010110111011
0101011000000100000000000010001010000000000000000010100010111111101001001000000000000
0001000000000001101100111011010000011001110011101110001100011001101011000001000000001
10001111011000000011011101111100101100100000110001000100000000010111100000111001000000
000001000010100110100101000000010000000000000111101110110000100111010000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000

```

Header fields of Ethernet

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Destination MAC Address	48 bits	e8:d8:d1:46:f3:f1	Receiver's MAC address
Source MAC Address	48 bits	00:1a:8c:6b:76:ac	Sender's MAC address
Type	16 bits	0x0800	0x800 indicates, Network Header type is IPv4 Header

Header fields of Network

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Version	4 bits	0x4	IP Datagram version 4
IHL	4 bites	0x5	5*32bits=160bits=20bytes
DSCP	6 bits	0xC50	---
ECN	2 bits	0x0	---
Total length	16 bits	0x28	Total length of 40 bytes

Identification	16 bits	0xbfd2	49106
flags	3 bits	0x2	2 bit More Fragment (MF)
Fragment offset	13 bits	0x8	This packet does not contain fragments.
Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Time to live	8 bits	0x80	128 Hops / Routers
Protocol	8 bits	0x0d	This packet should be give to TCP receive procedure. As its value indicates TCP.
Header checksum	16 bits	0x9da0	No checksum included in this header.
Source IP Address	32 bits	206.119.24.205	Source IP: 206.119.24.205, its local machine

Header fields of Transport Layer: ___TCP___

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Source Port	16 Bits	5580	Sender machine's application's logical port number 50547.
Destination Port	16 Bits	1773	Receiver machine's receiving logical port number 443 which indicates source wants communicate security using https protocol.
Sequence Number	32 Bits	0x666760299	Unique ID assigned by sender to maintain order of packers at receiver side.
Acknowledgement Number	32 Bits	0x3357567133	This is acknowledge of sent packet.
Header Length	4 Bits	0x4	Total header Length is 5*32bits=160bits=20bytes
Reserved Bits	6 Bits	0x0	-
URG	1 Bit	0	No URG
ACK	1 Bit	0	No ACK

PSH	1 Bit	0	No Push
RST	1 Bit	0	No RST
SYN	1 Bit	0	No SYN
FIN	1 Bit	0	NO Fin
Window Size	16 Bits	0x16427	4106
Checksum	16 Bits	0xfbb8	Error identification in packet.
Urgent Pointer	16 Bits	0x0	No urgent content in this packet.

Exercise-2: Input Sequence of UDP

```

11101000110110001101000101000110111100111111000100000000010100000101011010101011111100
100110011100001000000000000000000000000000000000000000000000000000000000000000000000000
000010000000000000000000000000000000000000000000000000000000000000000000000000000000000
001100011110110000000011011101100000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000

```

Header fields of Ethernet

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Destination MAC Address	48 bits	e8:d8:d1:46:f3:f1	Receiver's MAC address
Source MAC Address	48 bits	00:50:56:ab:f2:67	Sender's MAC address
Type	16 bits	0x0800	0x800 indicates, Network Header type is IPv4 Header

Header fields of Network

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Version	4 bits	0x4	IP Datagram version 4
IHL	4 bites	0x5	5*32bits=160bits=20bytes
DSCP	6 bits	0xC50	---
ECN	2 bits	0x0	---
Total length	16 bits	0x28	Total length of 40 bytes
Identification	16 bits	0x2159	8537
flags	3 bits	0x2	2 bit More Fragment (MF)
Fragment offset	13 bits	0x0	This packet does not contain fragments.

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Time to live	8 bits	0x80	128 Hops / Routers
Protocol	8 bits	0x11	This packet should be give to UDP receive procedure. As its value indicates UDP.
Header checksum	16 bits	0xa989	Validation disabled
Source IP Address	32 bits	172.16.11.71	Source IP: 172.16.11.71, its local machine

Header fields of Transport Layer: ____UDP____

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Source Port	16 Bits	443	Sender machine's application's logical port number 443.
Destination Port	16 Bits	80	Receiver machine's receiving logical port number 80 which indicates source wants communicate security using https protocol.
Checksum	16 Bits	0xb861	Its the checksum which is used for intergrity
Length	4 Bits	0x4	This means the length is of 20 bytes

[illegible]

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Destination MAC Address	48 bits	ff:ff:ff:ff:ff:ff	Receiver's MAC address
Source MAC Address	48 bits	e0:63:da:54:71:44	Sender's MAC address
Type	16 bits	0x0806	0x806 indicates, it is ARP Broadcast

Header field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Hardware type	16 bits	1	Using Ethernet 1
Protocol type	16 bits	0x0800	Using IPv4 protocol
Hardware size	8 bits	6	Hardware size of 6
Protocol size	8 bits	4	Protocol size of 4
Opcode	16 bits	1	Its is request
Sender MAC address	48 bits	e0:63:da:54:71:44	The mac of sender
Sender IP address	32 bits	172.16.8.33	The ip of sender
Target MAC address	48 bits	00:00:00:00:00:00	The mac of target
Target IP address	32 bits	172.16.7.157	The ip of target

Exercise-4: Input Sequence: ARP Reply

```

111000000110001111011010010101000111000101000100010011000001000110111111100111011111001
110001011000010000000011000000000000000010000100000000000000011000000100000000000000
100100110000010001101111111001110111110011100010111010110000010000000001110011101011100
000011000111101101001010100011100010100010010101100000100000000100000100001000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000

```

Header fields of Ethernet

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Destination MAC Address	48 bits	e0:63:da:54:71:44	Receiver's MAC address
Source MAC Address	48 bits	4c:11:bf:9d:f3:8b	Sender's MAC address
Type	16 bits	0x0806	0x806 indicates, it is ARP Reply

Header fields of _____ARP Reply_____

Header Field Name	Length of Field (in bits)	Header field Value (Hex Value)	Meaning
Hardware type	16 bits	1	Using Ethernet 1
Protocol type	16 bits	0x0800	Using IPv4 protocol
Hardware size	8 bits	6	Hardware size of 6
Protocol size	8 bits	4	Protocol size of 4
Opcode	16 bits	2	Its is reply
Sender MAC address	48 bits	4c:11:bf:9d:f3:8b	The mac of sender
Sender IP address	32 bits	172.16.7.58	The ip of sender
Target MAC address	48 bits	172.16.7.58	The mac of target
Target IP address	32 bits	172.16.8.33	The ip of target

CONCLUSIONS:

Post-Lab Assignment Questions**(Write answers in file pages, do not take print of this page)**

1. What do you mean by TTL (Time to Live)?
2. What is the significance of Sequence Number and Acknowledgment Number in TCP format?
3. What is the full form of the MAC address? What is the significance of source and destination MAC address?
4. What is the full form of IP, TCP, UDP and ARP?